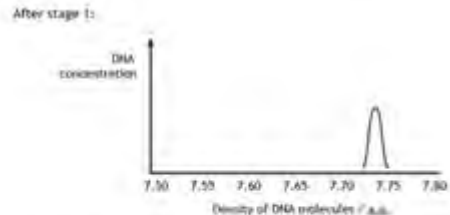
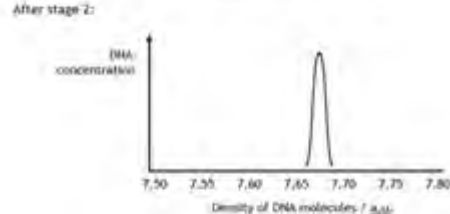
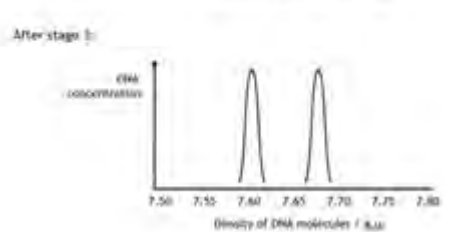


Question number	Answer	Additional guidance	Mark
5(a)	An answer that includes the following points: (semi-conservative) one {original / parent} strand retained (in each molecule) (1) (replication) more DNA (molecules) is produced / two new <u>molecules</u> made (1)	ACCEPT (in each new molecule) there is one old strand and one new strand NB both new molecules have one old strand and one new strand = 2 marks	(2)

Question number	Answer	Additional guidance	Mark
5(b)	A description that includes two of the following points: - adds nucleotides to each (new) DNA strand / {assembles / lines up} nucleotides (against the old strand / in new strand) (1) - forms phosphodiester bonds between (DNA) nucleotides (1) - repairs {damage / mistakes} in DNA (1)	ACCEPT description e.g. corrects wrong bases proofreading	(2)

Question number	Answer	Additional guidance	Mark
5(c)	An answer that includes the following points: after stage 1: - one peak at 7.74 (1) after stage 2 - one peak at 7.67 (1) - peak approx. twice height as after stage 1 peak (1) NB award if both peaks twice the height if two peaks shown after stage 3: - two peaks (1) - one at 7.60 and the other at 7.67 (1) NB award one of correct values plotted if only one peak has been shown correctly - both peaks approx. twice height as stage 1 (1) NB award if only one peak has been shown	IGNORE widths of curves After stage 1:  After stage 2:  After stage 3: 	(6)